

Step-1.5: Efficient Profile-Guided Pair-Adaptive Attribute Selection for LLM-Based Entity Resolution

1 Motivation

In Step-1, we reduce prompt cost by globally condensing tuples: pruning low-value attributes and compressing long fields. However, even among the retained attributes, not every field is equally important for every candidate pair. In some cases, an exact model number or DOI is already decisive. In others, the critical signal comes from an author mismatch, a year conflict, or a subtle title variation. Reusing the same condensed tuple for every pair therefore remains inefficient: it still spends tokens on attributes that are not decisive for that specific comparison, and it may distract the LLM from the small subset of fields that truly determine the judgment.

We therefore propose an intermediate step between global tuple condensation and full pairwise matching: **profile which attributes matter for representative matching and non-matching pairs, then transfer this attribute importance to new pairs by similarity.** The method has two stages. In the **profiling stage**, we select a representative set of labeled pairs and ask the LLM to identify the attributes that are most important for the decision. Each profiled pair becomes a compact *pair profile* containing its match label, its pairwise similarity pattern, and its important-attribute set. In the **matching stage**, we compare an unseen pair to the stored profiles, retrieve the most similar profiled pairs, and use their important-attribute sets to decide which attributes should be emphasized or retained for the final matching prompt.

The core idea is simple: instead of rediscovering attribute importance from scratch for every pair, we build a reusable library of profiled pair patterns and transfer them to similar cases. In other words, **Step-1 compresses tuples globally; Step-1.5 predicts which of the retained attributes matter locally for each pair.**

1.1 Illustrative Example

Consider the following profiled product pairs:

Profiled Match Pair:

Record A: [Title: Sony WH-1000XM4 Wireless Headphones] [Brand: Sony] [Model: WH-1000XM4] [Seller: TechDeals123]

Record B: [Title: Sony WH1000XM4 Noise Cancelling Headphones] [Brand: Sony] [Model: WH1000XM4] [Seller: AudioWorld]

LLM-Identified Important Attributes: Brand, Model, Title

Less Useful Attributes: Seller

Profiled Non-Match Pair:

Record C: [Title: Sony WH-1000XM5 Wireless Headphones] [Brand: Sony] [Model: WH-1000XM5]

Record D: [Title: Sony WH-1000XM4 Wireless Headphones] [Brand: Sony] [Model: WH-1000XM4]

LLM-Identified Important Attributes: Model, Title

Supportive but Non-Decisive: Brand

Now consider a new candidate pair:

New Pair:

Record E: [Title: Sony WH1000XM4 Headphones] [Brand: Sony] [Model: WH1000XM4] [Seller: BestAudio]

Record F: [Title: Sony WH-1000XM4 Wireless Noise Cancelling Headphones] [Brand: Sony] [Model: WH-1000XM4] [Seller: HeadphoneHub]

This new pair is much more similar to the profiled *match* pair than to the profiled *non-match* pair. Therefore, Step-1.5 transfers the important-attribute pattern {Brand, Model, Title} to this pair, while downweighting the seller field. Instead of passing every retained attribute with equal emphasis, the matcher now focuses on the fields that similar profiled pairs have already shown to be decisive.

2 The Profiling Strategies

Intuitively, we can consider the following strategies to make pairwise matching more adaptive after tuple condensation. The main goal is simple: identify only the most important retained attributes for each pair, then prioritize or restrict the prompt to those attributes. Think of it as finding the best balance between token efficiency, pair-specific focus, and matching quality.

Focusing on the two stages that deliver the most impact is a smart and practical choice. Here is the revised, streamlined section:

1. **Profiling Stage (Building a Library of Important-Attribute Patterns):** Select a representative set of matching and non-matching pairs, and ask the LLM to label which attributes were most important for the judgment.
 - **Keep:** Representative pairs that cover different similarity patterns, ambiguity levels, and pair types (e.g., easy matches, hard matches, obvious non-matches, confusing near-misses).
 - **Store:** For each profiled pair, keep the final match label, a compact pair signature (field-level similarity pattern), and the attributes that the LLM judged to be most important.
 - **Example (Profile Record):**
 - **Before (Raw Profile Pair):**
Record A: [Title: Efficient Data Integration with Large Language Models] [Authors: J. Smith; L. Wang] [Venue: VLDB] [Year: 2024] [DOI: 10.1000/xyz]
Record B: [Title: Efficient Data Integration using Large Language Models] [Authors: John Smith; Lei Wang] [Venue: Proc. VLDB Endow.] [Year: 2024] [DOI: 10.1000/xyz]
 - **After (Profiled Pair):**
[Pair Type: Match] [Important Attributes: Title, Authors, DOI] [Less Useful: Venue]
 - **Why:** A single profiling query converts one expensive LLM explanation into reusable attribute-selection knowledge. Instead of profiling every future pair from scratch, we can transfer this pattern to similar cases. **Impact: One-time profiling cost with broad reuse potential.**

2. **Similarity-Guided Attribute Transfer (Applying Profiles to New Pairs):** For a new pair, retrieve the most similar profiled pairs and use their important-attribute sets to decide which attributes should be emphasized in the final matching prompt.

- **How:** Represent the new pair using a field-level similarity signature (e.g., title similarity, author overlap, exact identifier match, year difference, missingness pattern). Retrieve the nearest profiled pairs and aggregate their important attributes by majority vote or similarity-weighted voting.
- **Example (Scholarly Record Pair):**
 - **Before (Condensed Pair):**
Record A: [Title: Efficient Data Integration with Large Language Models] [Authors: J. Smith; L. Wang] [Venue: VLDB] [Year: 2024] [Pages: 101–112] [DOI: 10.1000/xyz]
Record B: [Title: Efficient Data Integration using Large Language Models] [Authors: John Smith; Lei Wang] [Venue: Proc. VLDB Endow.] [Year: 2024] [Pages: 101–112] [DOI: 10.1000/xyz]
 - **After (Profile-Guided Selection):**
[Selected Important Attributes: Title, Authors, DOI]
[Downweighted Attributes: Venue, Pages]
 - **Why:** The matcher no longer treats all retained fields as equally important. It focuses on the small subset of attributes that similar profiled pairs have already shown to be decisive.
Impact: Medium-to-high token savings and sharper pair-specific reasoning.

3 Implementation Pathways

For this project, you are **encouraged to explore different techniques** for both the **profiling stage** and the **transferring stage**, and to compare how different design choices affect token efficiency, robustness, and matching quality.

3.1 Prototype-Oriented Exploration

A natural direction is to view the profiling stage as the construction of a small library of representative pair prototypes. Under this view, the main question is not how to build one exact prototype set, but how different kinds of representative profiles influence downstream attribute transfer.

For the **profiling stage**, you are encouraged to explore different techniques for choosing representative matching and non-matching pairs, defining what makes a pair “representative,” and deciding how much information should be stored in each profile. For the **transferring stage**, you are encouraged to explore different techniques for relating a new pair to one or more prototypes, and for deciding whether the important attributes from those prototypes should be transferred directly, combined, or softened.

3.2 Similarity- or Signature-Oriented Exploration

A second direction is to organize profiled pairs around pairwise similarity patterns or pair signatures. In this view, the central issue is how to describe the relationship between two records in a way that is informative enough to support attribute transfer, yet compact enough to remain efficient.

For the **profiling stage**, you are encouraged to explore different techniques for characterizing pairs, grouping similar pair patterns, and deciding whether profiles should be organized globally

or separated into different match and non-match regions. For the **transferring stage**, you are encouraged to explore different techniques for retrieving, aggregating, or comparing similar profiles when deciding which attributes matter for a new pair.

3.3 Learning-Based or Hybrid Exploration

A third direction is to treat profiled pairs as supervision for a learned or semi-learned attribute selector. Rather than relying only on direct profile retrieval, the system may learn broader regularities about when certain attributes become decisive.

For the **profiling stage**, you are encouraged to explore different techniques for turning LLM-labeled important attributes into reusable supervision signals, weak rules, or profile summaries. For the **transferring stage**, you are encouraged to explore different techniques that predict important attributes directly, combine prediction with profile retrieval, or use hybrid strategies that adapt to confidence or ambiguity.

This direction has the highest exploratory upside, because it asks whether the knowledge extracted during profiling can generalize beyond the exact profiled pairs. It also opens up a broader research question: *is attribute importance better transferred by retrieving similar profiles, by learning a predictor, or by combining both?*

4 Project Timeline and Deliverables

The total timeline for this project is expected to be **3 to 4 weeks**. As research students, you will be driving this project through two main phases.

Phase 1: Exploration and Solution Design (1–2 Weeks)

Please select one of the three directions outlined above that you feel is the most promising. During this initial phase, you should:

- Explore possible solutions and identify specific, promising techniques within your chosen pathway.
- Discuss and formulate your proposed methodology.
- Thoroughly justify your solution, providing a clear argument for exactly why you think your chosen approach will successfully address the problem (i.e., balancing token efficiency and matching F1-score).

Phase 2: Implementation and Experimentation (1–2 Weeks)

After proposing your solution, we will review it together. Once you revise your approach based on the provided feedback, you will transition into the execution phase. During this time, you should:

- Implement your Step-1.5 pipeline in code.
- Run experiments on standard datasets to empirically validate your approach against appropriate baselines.
- Document your findings, focusing on token savings, F1-score retention, and the robustness of your attribute selection.